INTERNSHIP REPORT

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Name of Host Organization: NATIONAL INSTITUTE OF TECHNOLOGY CALICUT
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Declaration

I hereby declare that the internship report titled "Lightweight Transformer Equalizer for Doppler-Affected Rayleigh Environments under Low Frequency Constraint" is a genuine record of the work carried out by me during the internship under the guidance of DR. SUDEEP P.V., and that this report has not been submitted elsewhere for academic credit or professional evaluation.

Signature of Student
Date:

Certificate

This is to certify that Jeswin Antony Chungath, an Electronics and Communication Engineering student from Government Engineering College Thrissur, has successfully completed an internship at National Institute of Technology Calicut from 20 May 2025 to 3 August 2025 under the guidance of Dr. Sudeep P. V. The intern has shown sincere effort and involvement in the assigned tasks

Seal & Signature
(With Date, Name, Designation)

Acknowledgements

I would like to express my sincere gratitude to the IEEE Computational Intelligence Society (CIS) Joint Chapter for facilitating this valuable internship opportunity. I am deeply thankful to my home institution, Government Engineering College Thrissur, for their continuous academic support and encouragement. My heartfelt appreciation goes to the National Institute of Technology Calicut, the host organization, for providing a stimulating research environment and access to excellent resources. I am especially grateful to Dr. Sudeep P. V., my internship guide, for his insightful mentorship, technical guidance, and unwavering support throughout the internship period.

1 Introduction

We have come to an era of rapidly developing infrastructure, where buildings and obstacles pop out every day. This has led to ever developing requirement for non line of sight communication (NLOS). Also present requirements for high bandwidth communication mid traveling has added complexity to this problem. Mobile-Mobile communication, urban cellular networks, indoor wireless systems. This has lead to a situation where there is no dominant line of sight communication path, only scattered multipath components.

This has led to a prevailing problem of doppler spread. Doppler spread refers to the range of frequency shifts caused by the relative velocities of multiple scatterers in a multipath environment. Unlike a single Doppler shift (which is a fixed offset), Doppler spread captures the variation in frequency shifts across all paths. It is a measure of channel time variability. Low Doppler Spread means slow changing or quasi-static channel, whereas High doppler spread would mean a fast fading channel [1].

The Transformer is a deep learning architecture introduced by Vaswani et al. in 2017 [2], which has since become foundational in the field of natural language processing. Unlike traditional recurrent models, the Transformer relies entirely on a self-attention mechanism to model dependencies between input elements, enabling parallel processing of sequences and improved scalability. Its encoder–decoder structure, combined with positional encoding and multi-head attention, allows it to capture complex contextual relationships across long sequences. This architecture has led to significant advancements in language modeling and has become the basis for state-of-the-art models such as BERT, GPT, and T5. While Transformer models are generally data and power hungry, through this project I aim to achieve significantly lower computation power consuming equalizer under low frequency restriction on carrier frequency.

2 Organization Profile

This project is focused on developing a ML model for dealing with Rayleigh effects occurring in wireless communications[3]. A python program is used to simulate the Rayleigh channel effects, which includes AWGN, doppler spread, fading. Libraries PyTorch, Numpy, Scipy and os are used for python simulation.

3 Internship Objectives

This study is aimed at creating a Non-Adaptive ML model robust enough to deal with multipath effects and doppler spread by using a ML Model. The program has a model which will act as an equalizer to the distorted channel output and equalize the result to obtain an approximate version on the original signal. Then it is passed on for classification which will revert it back to original bits. The ML model used here is a Transformer [2].

To achieve this, the study focuses on constructing a Transformer-based ML model that functions as a channel equalizer. The model receives the distorted output from the communication channel typically affected by inter-symbol interference (ISI), fading, and frequency shifts and learns to reconstruct a signal that closely approximates the original transmitted waveform. Unlike traditional adaptive equalizers [4], this approach leverages the Transformer's attention mechanism to capture long-range dependencies and temporal correlations in the signal, enabling effective compensation for complex channel impairments [5].

Once the equalized signal is obtained, it is forwarded to a classification module that maps the reconstructed waveform back to its corresponding bit sequence. This step completes the end-to-end recovery pipeline, allowing the system to infer the original transmitted data with high fidelity. The classification process is tightly integrated with the equalization stage, ensuring that the model learns a joint representation optimized for both signal restoration and bit-level accuracy [6].

The overarching aim is to demonstrate that a non-adaptive, data-driven approach using Transformers can outperform or complement conventional signal processing techniques in challenging wireless environments [7].

4 Work Description

4.1 System Implementation

A Python Program is used to simulate the Rayleigh Channel. A data set of bits is generated using which is split for training and testing. The transformer model is trained over multiple epochs and then tested for performance. Here a transformer model is used as an equalizer. The Short Time Fourier Transform (STFT) is used to extract the frequency feature of the signal, extracted for a specific window and then passed onto the transformer model along with time, magnitude, phase information. The Transformer processes the signal and produces an output vector which is then passed on for classification[8]. The code is available at `https://github.com/JeswinAnto/Transformer-to-combat-Rayleigh-/tree/main`.

4.2 Training Strategy

The model is trained using Cross Entropy Loss Function with Loss as a function of the BER. The model is trained over multiple epochs, with early stopping enabled to prevent performance from degrading. The training data is generated is using a random number generator in python. The model is then tested over a separate data set so as to detect overfitting. The constellation diagram for QPSK for initial transmitted bits, then for distorted channel output and then detected output after equalization is plotted.

4.3 Performance Metrics

The performance metrics used here are Symbol Error Rate (SER), Bit Error Rate (BER). Also a comparison across various models[9]

5 Learning Outcomes

Transformer

Understood the intricate mechanisms behind the transformer. Like Input embedding's capability to capture semantic, contextual relationships with higher dimensional learnable embedding.

The Positional encoding vectors to carry long range and short range dependencies. The ability of Attention mechanisms to capture non-linear complex relationships with linear pre-processing of inputs into Query, Key and Value followed up by extraction of features from inner products of generated vectors in multiple higher dimensional subspace.

Which is followed up by Layer Normalization for stability and residual addition to deal with degradation of gradients. Then followed by a FeedForward Network to increase the expressiveness of the information.

A ReLU unit is put between to introduce sparsity and add non-linearity to the system. The layers are repeated to enhance he depth of the transformer [2].

Equalizer

It acts as a channel inversion unit. Through Learned information upon training on a sample data an appropriate strategy is adopted to minimize the losses incurred. Here the ability of transformer model to capture non-linear shift variant relationships is leveraged to create a Learned machine to invert the effects of Rayleigh distortion.

The Input vector carries the higher dimensional vector of each distorted output along with STFT capturing short range Frequency resolutions with sufficient hop size for maintaining sufficient overlap, and with Magnitude, Phase bins and time encoding. This vector is feature extracted to the Input Embedding of the transformer.

Followed by Processing with cross entropy loss function. The result of the training which gives the equalizer a transfer function equivalent to the inverted channel transfer response [5].

6 Challenges Faced & Solutions

Bottleneck on Learning

Under simple transmission of Rayleigh-distorted signal sequences, the model's performance stagnates at 50% Bit Error Rate (BER), which corresponds to the probability of random guessing. Despite enhanced model dimensions and early stopping, the best outcome remains at 50% BER. Even with added complexity and CNN-based pre-processing, the model fails to learn the channel behavior.

One contributing factor may be the use of QPSK modulation, which is less resilient to time-varying frequency shifts compared to more robust schemes like MFSK and OFDM [9].

Overfitting

The model consistently overfits when exposed to complex time-variant properties of the distorted signal. While it performs well on training data, it fails significantly on testing data. Increasing model dimensions accelerates overfitting rather than improving generalization.

Complex Model Parameter–Performance Relationships

There is no straightforward relationship between model parameters and performance. Adjusting complexity either increasing or decreasing does not yield predictable improvements or degradations. This variability also extends to data pre-processing, complicating the design of robust and stable model configurations.

Computation Cost

Transformer-based models are computationally intensive, consuming significant resources without delivering proportional performance gains in this context. Training more complex models often becomes impractical due to resource constraints.

Low Frequency Constraint

The program achieves QPSK Modulation on a low frequency carrier frequency of 2MHz as compared to RF frequencies of 100's of MHz to GHz range used in conventional system. This significantly affects the model performance under heavy doppler spread conditions like 50KHz for LEO Sattelites.

Solution

To address learning bottlenecks, frequency, phase, time, and magnitude information were explicitly packaged and fed to the model, rather than relying solely on model complexity to extract these features. To balance this added pre-processing, model parameters were significantly reduced:

Original:	Input_dim = 64,	num_heads = 4,	num_layers = 3,	feedforward_dim = 256
Reduced:	Input_dim = 16,	num_heads = 1,	num_layers = 1,	feedforward_dim = 64

This adjustment resulted in an atleast 8 times improvement and at best the BER was zero BER and Symbol Error Rate (SER). Additionally, early stopping was enabled with sufficient patience (relative to the base number of epochs) to mitigate overfitting [10].

7 Contribution to Host Organization

Developed Reusable Transformer-Based Equalizer Framework

Implemented a modular Transformer architecture for channel equalization in communication systems. The design supports various signal formats and channel models, making it adaptable for future research and deployment. Key features include attention mechanisms for temporal modeling and configurable layers for experimentation [2].

Built Feature Extraction Pipeline

Created a pre-processing pipeline to convert raw channel outputs into structured inputs for ML models. This includes magnitude, phase, and frequency binning, which improve learning efficiency and model performance [8].

Established Benchmarking Methodology

Defined a consistent evaluation strategy using metrics like Bit Error Rate (BER), Signal-to-Noise Ratio (SNR), and inference time. The methodology enables fair comparison between ML-based equalizers and traditional approaches across different channel conditions [6].

Documented Implementation for Future Research

Compiled detailed documentation covering model architecture, training procedures, and configuration settings. Included usage guidelines and troubleshooting notes to support reproducibility and ease of adoption by other researchers.

8 Conclusion & Future Scope

A Transformer based Equalizer was created in python, simulated on training, testing and validation. Significant reduction in BER, SER was obtained for this model compared to Transformer Model. The model performance is exceptional in learning and attaining significant accuracy at very short interval of training, 10 epochs in his case which is very short compared to standard of 50 epochs. Within this short period the model has managed to get a BER consistently zero for doppler frequencies 50, 500 Hz (as of Validation), even in case of extremely heavy distortion of 50KHz which is usually seen in LEO sattelites the model is able to achieve 2-3% error. While in case of Test BER the model BER is degraded yet it retains high accuracy. This is established in a ultra small scale transformer model which utilizes nearly 3000 neurons only as compared to 20k-200k neurons in BiLSTM and 10k-100k in FCNN. The table showing Intra-Training Validation and Final Testing BER is at the last page of appendix.

- **Transfer Learning for Modulation Schemes**
  The trained model weights can be repurposed as a pre-trained baseline for similar applications involving modulation schemes such as ASK, FSK, and QAM, particularly in scenarios affected by Doppler spread.

- **Dynamic Model Simplification Using Advanced ML Architectures**
  Emerging machine learning models like *Mixture of Recursions (MOR)* offer promising avenues for dynamically reducing model complexity. This can lead to significant improvements in power efficiency, computational time, and resource utilization.

- **Learned Pre-Processing for Hardware Optimization**
  Traditional pre-processing steps such as *Short-Time Fourier Transform (STFT)* and *binning* can be replaced by a learned ML model. This approach enables direct

integration into the system, thereby enhancing performance in real-time hardware implementations [7].

9 References

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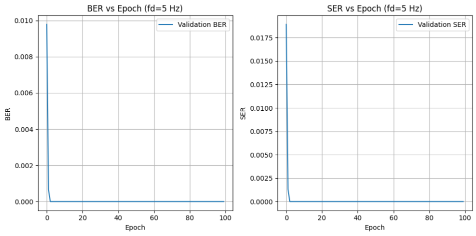
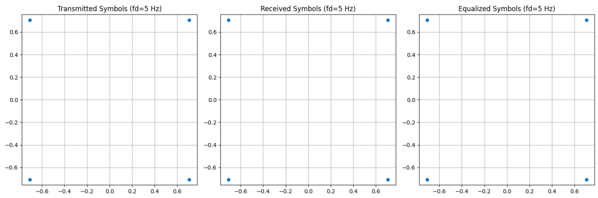
10 Appendix

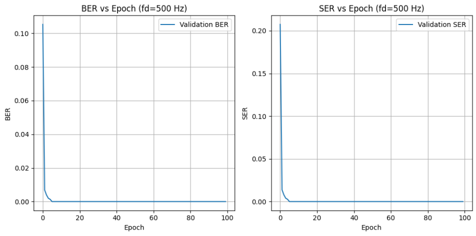


(a) Validation BER, SER vs Epoch (fd = 5 Hz)

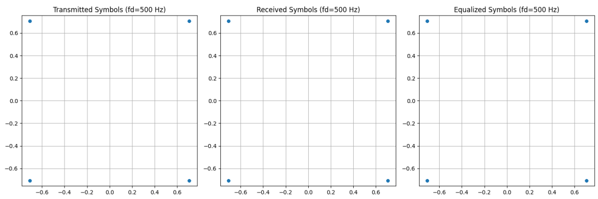


(b) Constellation Diagram (fd = 5 Hz)

Figure 1: Performance and Symbol Constellation at Doppler Frequency of 5 Hz

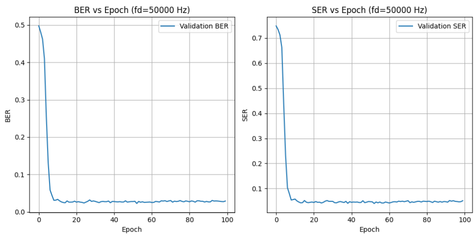


(a) Validation BER, SER vs Epoch (fd = 50 Hz)

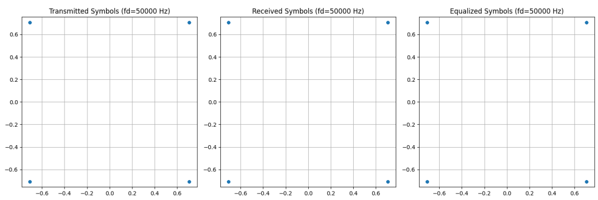


(b) Constellation Diagram (fd = 500 Hz)

Figure 2: Performance and Symbol Constellation at Doppler Frequency of 50 Hz



(a) Validation BER, SER vs Epoch (fd = 50000 Hz)



(b) Constellation Diagram (fd = 100 Hz)

Figure 3: Performance and Symbol Constellation at Doppler Frequency of 100 Hz

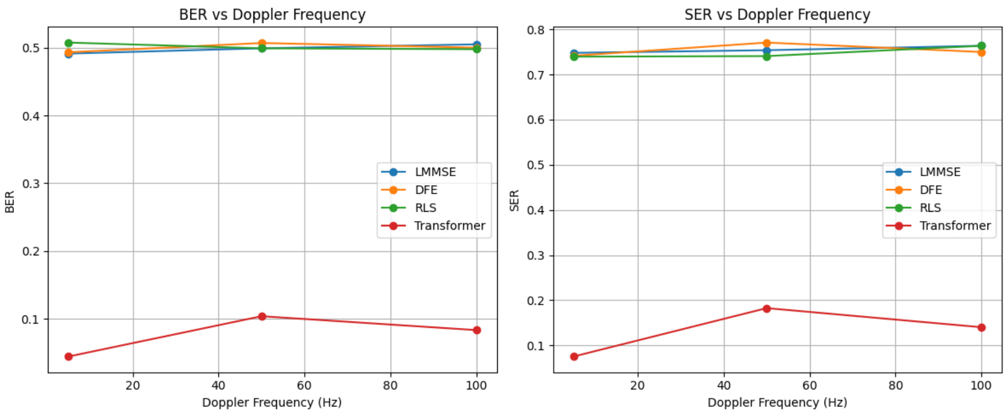


Figure 4: Performance Comparison of Transformer to earlier Equalizers

Table 1: Training and Test Metrics for Transformer-Based Equalizer Across Different Doppler Frequencies. Note: The carrier frequency is configured as 2000000 Hz (interpreted as 2 MHz) in the simulation.

Epoch	**Train Loss**	**Val Loss**	**Val BER**	**Val SER**
Doppler Frequency: 5 Hz				
0	1.107696	0.300212	0.009785	0.018917
10	0.003065	0.000864	0.000000	0.000000
20	0.021948	0.000244	0.000000	0.000000
30	0.000469	0.000066	0.000000	0.000000
40	0.000198	0.000024	0.000000	0.000000
50	0.000385	0.000021	0.000000	0.000000
60	0.000434	0.000013	0.000000	0.000000
70	0.000041	0.000005	0.000000	0.000000
80	0.000027	0.000002	0.000000	0.000000
90	0.000075	0.000004	0.000000	0.000000
Final Test BER: 0.000000, Final Test SER: 0.000000				

Epoch	**Train Loss**	**Val Loss**	**Val BER**	**Val SER**
Doppler Frequency: 500 Hz				
0	1.310929	0.666525	0.105349	0.207436
10	0.121498	0.003874	0.000000	0.000000
20	0.031843	0.000997	0.000000	0.000000
30	0.024577	0.000495	0.000000	0.000000
40	0.012873	0.000163	0.000000	0.000000
50	0.013445	0.000120	0.000000	0.000000
60	0.066487	0.000126	0.000000	0.000000
70	0.004787	0.000036	0.000000	0.000000
80	0.022141	0.000071	0.000000	0.000000
90	0.023105	0.000070	0.000000	0.000000
Final Test BER: 0.049708, Final Test SER: 0.087719				

Epoch	**Train Loss**	**Val Loss**	**Val BER**	**Val SER**
Doppler Frequency: 50,000 Hz				
0	1.462988	1.413531	0.497391	0.748206
10	0.199115	0.165929	0.033594	0.058056
20	0.135004	0.116192	0.026093	0.044357
30	0.112584	0.121152	0.028050	0.048924
40	0.095633	0.134363	0.027723	0.045010
50	0.082487	0.141493	0.027723	0.047619
60	0.082371	0.136120	0.025114	0.042401
70	0.075511	0.152296	0.030333	0.051533
80	0.065058	0.158896	0.027071	0.048271
90	0.071508	0.150266	0.026745	0.047619
Final Test BER: 0.046784, Final Test SER: 0.087070				

Doppler Frequency (Hz)	**Test BER**	**Test SER**
5	0.000000	0.000000
500	0.049708	0.087719
50,000	0.046784	0.087070